CODEA VR: An Immersive Prototype for Teaching Computational Thinking in a Rural School

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Abstract

Within the CODEA community outreach project, a Meta Quest 3S-based virtual reality prototype named **CODEA VR** was developed to support computational-thinking activities through block-based challenges in a 3D environment. The study addresses the limited integration of structured programming-related activities in Ecuadorian secondary education, particularly in a rural school context with limited access to technological resources, connectivity, and specialized digital training. An exploratory evaluation was conducted at the Yachay Millennium Educational Unit (UEMY), a rural educational institution in Urcuquí, Ecuador, with students from Upper Basic General Education and Unified General Baccalaureate. The analysis focuses on questionnaire data related to digital competence, user engagement, and user experience.

Literature Review

Computational thinking is recognized as a transversal competence related to problem formulation, solution design, sequencing, and evaluation. Block-based programming supports novice learners by reducing syntax barriers and allowing them to focus on logic, sequencing, debugging, and problem-solving [1]. Virtual reality can enrich this process through immersive, interactive, and feedback-based learning experiences [2].

Research Gap

In Ecuadorian secondary education, programming-related learning activities are not consistently integrated as a continuous pathway across Upper Basic General Education and Unified General Baccalaureate. This gap is more evident in rural school contexts, where access to technological resources, connectivity, and specialized digital training is often limited. Therefore, there is a need to explore complementary tools such as **CODEA VR**, developed within the CODEA project, to support computational-thinking activities through immersive and interactive learning experiences.

Research Objectives

The present study focuses on the following objectives:

- **Objective 1:** Design an immersive virtual reality prototype, named **CODEA VR**, using Meta Quest 3S to support the development of computational thinking through interactive learning experiences.
- **Objective 2:** Develop and implement a functional prototype that enables students to construct, execute, and reflect on problem-solving strategies using block-based programming and real-time feedback in a 3D environment.
- **Objective 3:** Evaluate, in a rural school context (UEMY), students' digital competence, user engagement, and user experience during their interaction with **CODEA VR**, to identify strengths and improvement opportunities for future design iterations.

Study Methodology

The study followed a structured methodology aligned with the defined research objectives:

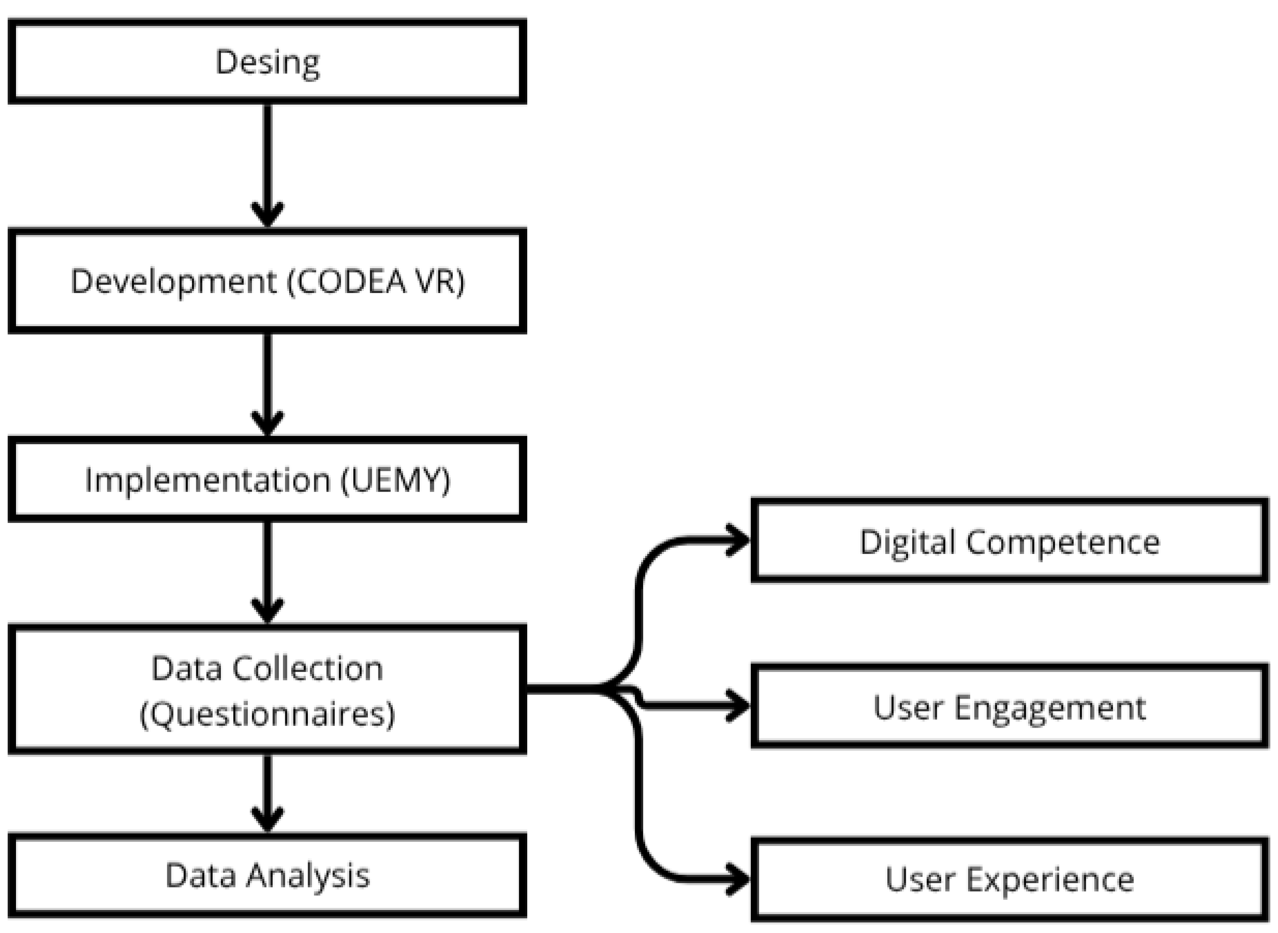


Figure 1. Methodological flow for the design, implementation, and exploratory evaluation of **CODEA VR**.

CODEA VR Interface and Educational Context

CODEA VR allows students to solve computational-thinking challenges by arranging visual blocks that control a robot in a 3D environment. Students select instructions, build a sequence, execute the robot's movement, and receive immediate feedback.

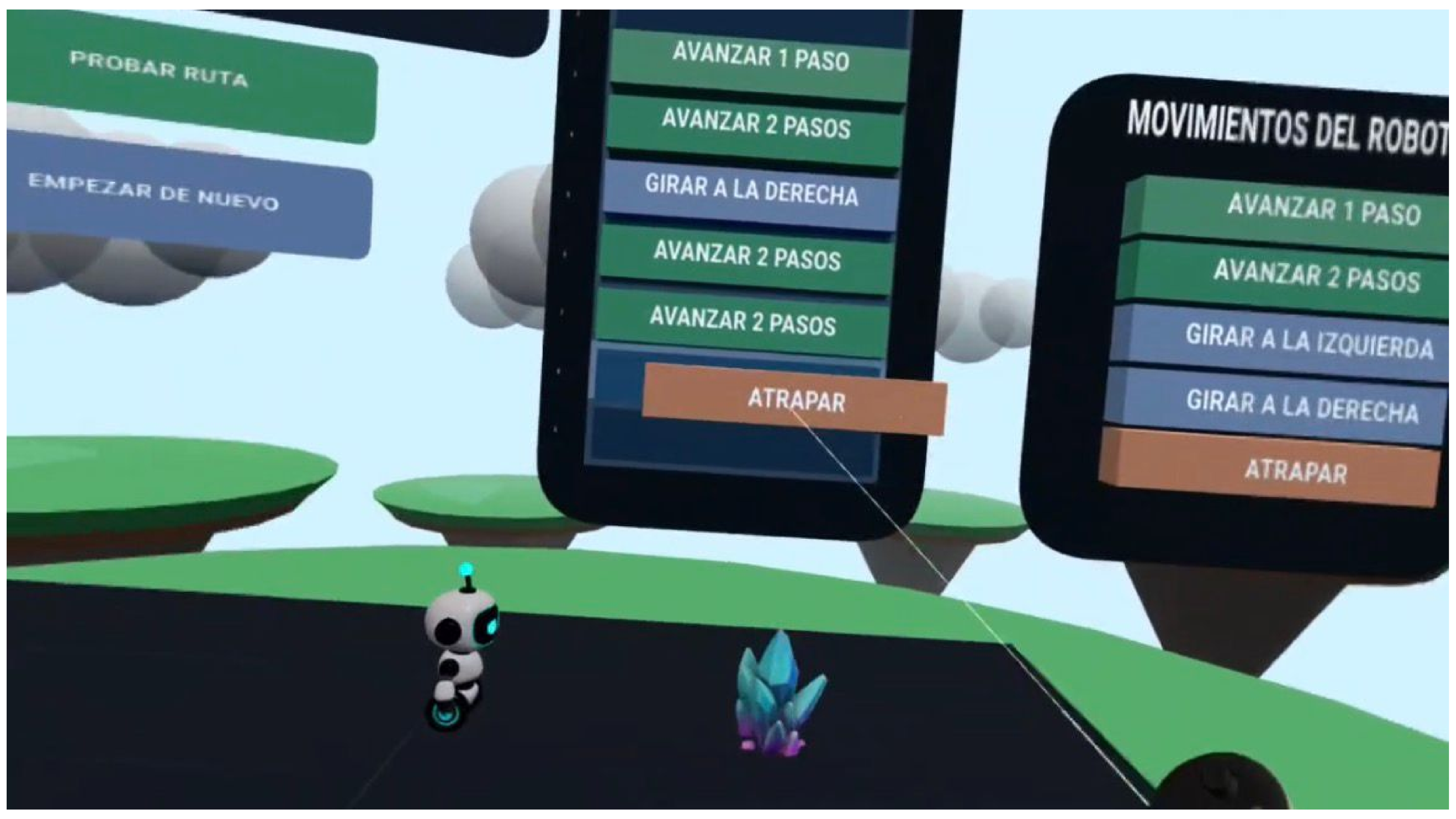


Figure 2. **CODEA VR** interface showing the robot, visual instructions, and 3D challenge environment.

Sample Description

The study was conducted with students from a single rural educational institution (UEMY), including participants from Upper Basic General Education (EGB) and Unified General Baccalaureate (BGU).

- **Sample size:** 93 students.
- **Age range:** 11–18 years.
- **Gender distribution:**
  - Female: 47 students
  - Male: 46 students
- **School levels:**
  - 8th EGB: 23 students
  - 9th EGB: 12 students
  - 10th EGB: 23 students
  - 1st BGU: 11 students
  - 2nd BGU: 12 students
  - 3rd BGU: 12 students

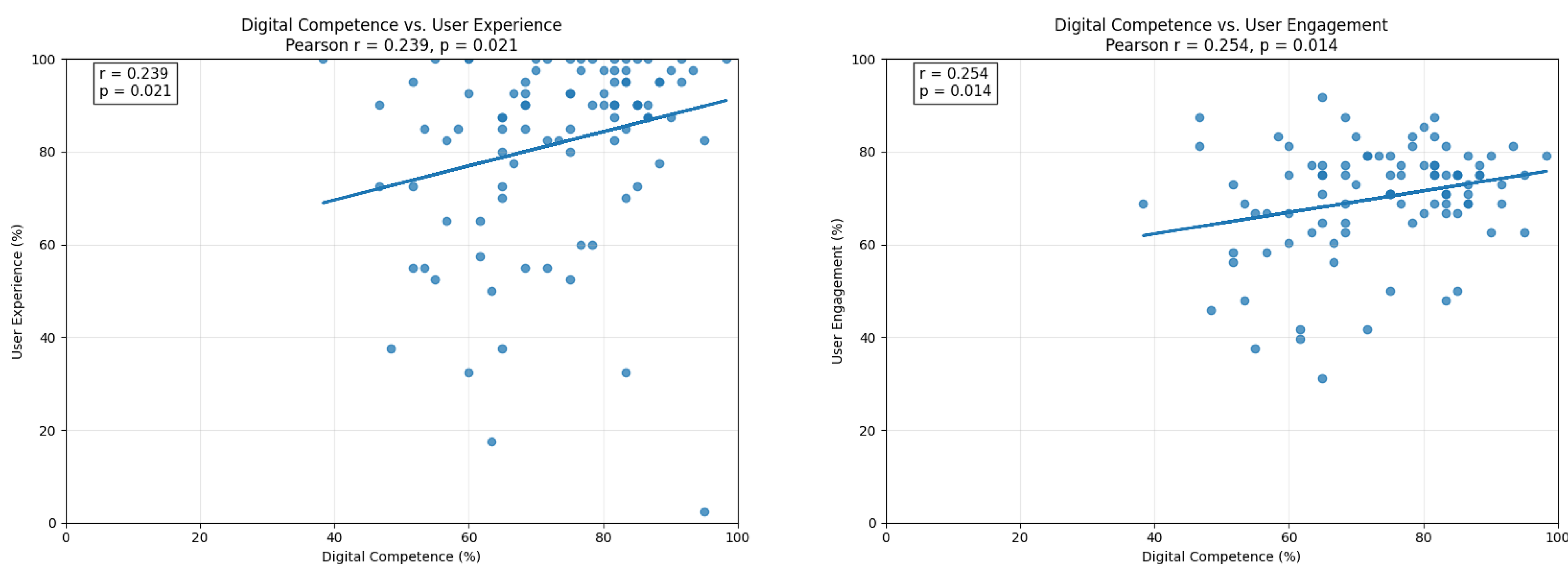
Data were collected through questionnaires designed to evaluate three main dimensions: **digital competence**, **user engagement**, and **user experience**.

Figure 3. Student using Meta Quest 3S during the **CODEA VR** evaluation.

Results and Discussion

The analysis shows a **slight but consistent positive relationship** between students' digital competence and their experience and engagement when using **CODEA VR**.

- **Digital competence vs. user experience:** a weak positive correlation ($r = 0.239$, $p = 0.021$).
- **Digital competence vs. user engagement:** a weak positive correlation ($r = 0.254$, $p = 0.014$).



A. Digital competence vs. user experience **B.** Digital competence vs. user engagement

Although the correlations are low, they are statistically reliable ($p < 0.05$), suggesting that students with higher digital skills tend to report a slightly better experience and engagement. However, these results also suggest that **usability, feedback, and challenge design** play a more important role in shaping students' interaction with **CODEA VR**, especially in a rural educational context.

Conclusions

- **CODEA VR** shows potential as an immersive tool to introduce computational thinking in a rural school context.
- Students were able to engage with block-based challenges using Meta Quest 3S, despite different levels of digital competence.
- The correlation graphs showed weak but statistically significant positive relationships between digital competence and user experience (r = 0.239, p = 0.021), and between digital competence and user engagement (r = 0.254, p = 0.014).
- Future iterations should improve usability, instructions, feedback, and challenge progression.

Acknowledgment

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Selected references

1. Wing, J. M. (2006). Computational thinking. *Communications of the ACM*, 49(3), 33–35.
2. Conrad, M., Kablitz, D., & Schumann, S. (2024). Learning effectiveness of immersive virtual reality in education and training: A systematic review of findings. *Computers and Education: X Reality*, 4, 100053.

CODEA VR Demo

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